

# Manufacturing Engineering Technology I

TEKS §127.813 (working draft) · CTE Level 2 · Grade 10 · Mr. Gavaldon · Da Vinci School

**Week 2**
**MONDAY Day 6 - Measurement & Tools**

Aug 17

**OBJECTIVE**

Students will read a metric vernier caliper to a least count of 0.02 mm, distinguish accuracy, precision, and resolution using trial data (mean and range), and apply a bilateral tolerance to accept or reject a part.

**TEKS**

§127.813 (draft, verify strand) measurement, tools & precision in manufacturing

**ELPS / EB SUPPORT**

ELPS/EB supports at-level: pre-teach accuracy/precision/tolerance with a picture card (PV, DP); model the vernier reading step-by-step and repeat slowly (RS, VC); hands-on measuring stations with a partner and native-language number talk (PN, CD); wait time on the classify-the-data exit question (WT).

**AGENDA • 45 MIN**

- 6' Bell-work: estimate this block's thickness by eye, then we will measure it
- 10' Precision vs. accuracy vs. resolution; reading a vernier caliper to 0.02 mm
- 8' Worked examples: vernier reading + tolerance limits (GO / NO-GO)
- 16' Measuring lab: 3 objects x 3 trials, compute mean and range, check tolerance
- 5' Exit ticket: classify a data set as accurate and/or precise

**SLIDES / ACTIVITY**

Measure Like a Machinist: Precision, Tolerance & the Vernier Caliper

**NOTEBOOK**

Record your station measurements. Define precision vs. tolerance in your own words.

**TURN IN**

Precision, Tolerance & the Vernier Caliper - Lab + Problem Set

**TUESDAY Day 7 - Manufacturing Math**

Aug 18

**OBJECTIVE**

Students will convert between inches (fractional and decimal) and millimeters using the exact factor 1 in = 25.4 mm, compute tolerance limits and a worst-case tolerance stack-up, and evaluate a clearance fit - showing all work to the correct resolution.

**TEKS**

§127.813 (draft, verify strand) applied mathematics & measurement in manufacturing

**ELPS / EB SUPPORT**

ELPS/EB supports at-level: keep a conversion reference card (1 in = 25.4 mm) and worked-example frames (CD, RS); pre-teach nominal / tolerance / limits / stack-up (PV); dual-language number talk with a partner (PN); the mm = 25.4 x in graph is a visual anchor for the relationship (VC, DP); extra time on the stack-up and fit problems (ET).

**AGENDA • 45 MIN**

- 6' Bell-work: convert 2.5 in to mm (show the setup)
- 12' Conversions (1 in = 25.4 mm), fraction<->decimal, tolerance limits and stack-up
- 8' Worked examples: convert + tolerance range + clearance fit
- 14' Practice set: real part dimensions (show all work)
- 5' Go over answers + exit ticket

**SLIDES / ACTIVITY**

The Math Behind the Part: Conversion, Tolerance & Fit

**NOTEBOOK**

Keep the conversion table. Work 2 example problems with all steps shown.

**TURN IN**

Manufacturing Math - Conversion, Tolerance & Fit Problem Set

## OBJECTIVE

Students will define user parameters and driving relations (expressions) in Fusion 360, apply geometric and dimensional constraints to fully constrain a sketch, and explain how design intent lets one parameter change regenerate a manufacturable part correctly.

## TEKS

§127.813 (draft, verify strand) parametric CAD for manufacturable parts

## ELPS / EB SUPPORT

ELPS/EB supports at-level: build steps mirrored on-screen and posted as Classroom screenshots (VC, DP); pre-teach parameter / expression / constraint / fully-constrained (PV); rephrase and slow each build step (RS); finishers pair with peers who are stuck (PN); extra time to reach a fully-constrained sketch (ET).

## AGENDA • 45 MIN

6' Bell-work: change one number and the whole part updates - why?

6' Watch: Fusion first look - dimensions drive the shape

10' Parameters, expressions/relations, constraints and fully-constrained sketches

18' Guided build: parametric plate driven by width / height / thickness / edge

5' Exit: list your 3 parameters and what each controls; save

## SLIDES / ACTIVITY

Parametric Design: Parameters, Relations & Constraints

## NOTEBOOK

Write the 3 parameters you set (name = value) and what each controls on the part.

## TURN IN

Parametric Design - Parameters, Relations & Constraints

## OBJECTIVE

Students will write a formal problem statement for a mounting plate, select a clearance hole from a fastener chart, compute plate size, hole coordinates, and an edge-distance check from a component's bolt pattern, and begin modeling the plate parametrically in Fusion 360.

## TEKS

§127.813 (draft, verify strand) apply parametric CAD to a defined part; documentation

## ELPS / EB SUPPORT

ELPS/EB supports at-level: the part-planning sheet organizes the math before CAD (CD, DP); a reference model and a dimensioned sketch are shown on screen (VC); pre-teach clearance hole / edge distance / datum / fastener designation (PV); partner talk in native language while planning (PN); extra time to finish the sketch (ET).

## AGENDA • 45 MIN

5' Bell-work: name 2 components a plate might hold, and the fastener size

12' Project brief: problem statement, clearance holes, edge distance, datums, tolerances

8' Worked example: size a plate from a component's bolt pattern

15' Plan YOUR dimensions on the sheet, then begin modeling in Fusion

5' Save (yourname-plate) + exit

## SLIDES / ACTIVITY

Project: Mounting Plate - From Brief to Parametric Model

## NOTEBOOK

Record your plate dimensions and hole sizes. Sketch the plate with dimensions labeled.

## TURN IN

Part-Planning Sheet - Mounting Plate (with calculations)

## OBJECTIVE

Students will map the Autodesk Certified User (Fusion) exam domains to features of their mounting plate, run a parametric robustness test (change a parameter, predict and verify the update), and complete an evidence-based peer review of a partner's model against a rubric.

## TEKS

§127.813 (draft, verify strand) industry certification pathway; collaboration

## ELPS / EB SUPPORT

ELPS/EB supports at-level: the certification path is shown as a simple roadmap graphic (VC, DP); pre-teach certification / domain / rubric / performance-based (PV); peer-review pairs an EB student with a helper and native-language discussion is welcome (PN); rephrase and slow the test-strategy tips (RS); wait time on the self-check (WT).

## AGENDA • 45 MIN

5' Bell-work: what does an industry certification get you?

10' Autodesk Certified User: logistics + the 5 objective domains

8' Worked: parametric robustness test + how a failure shows

17' Continue modeling your plate; run the robustness test; peer-review a partner

5' Self-check + exit

## SLIDES / ACTIVITY

Certification Kickoff + Parametric Robustness & Peer Review

## NOTEBOOK

List the certification domains. Note where you are on your plate and one thing to finish Monday.

## TURN IN

Certification Domains, Robustness Test & Peer Review